

MATH 189 HW 5

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Question 1

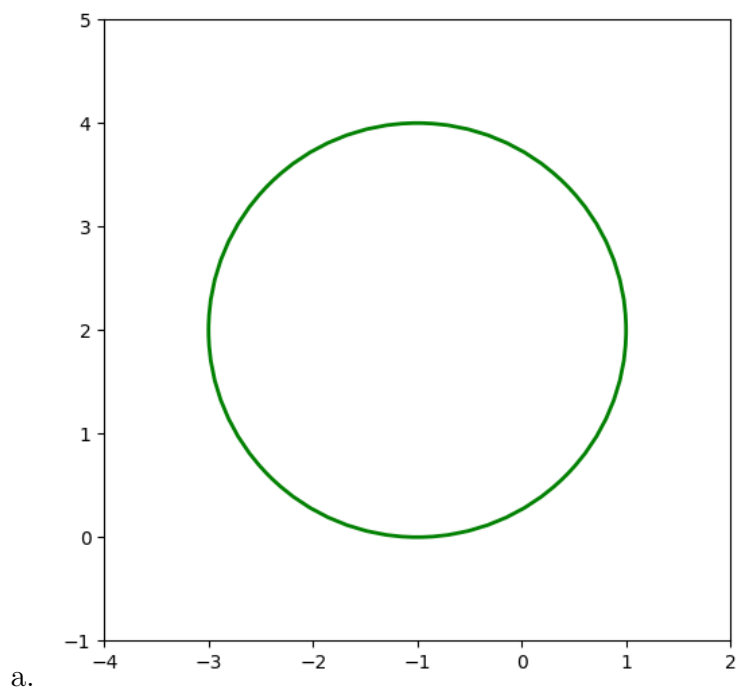
Since boosting takes linear combinations of depth-one trees, which are univariate functions, we get the sum of univariate functions, which is just an additive model.

Question 2

We split each cell by finding the predictor X_j and cutpoint s that minimizes

$$\sum_{i: X_{ij} < s} (Y_i - \hat{Y}_{R_1})^2 + \sum_{i: X_{ij} \geq s} (Y_i - \hat{Y}_{R_2})^2$$

Question 3

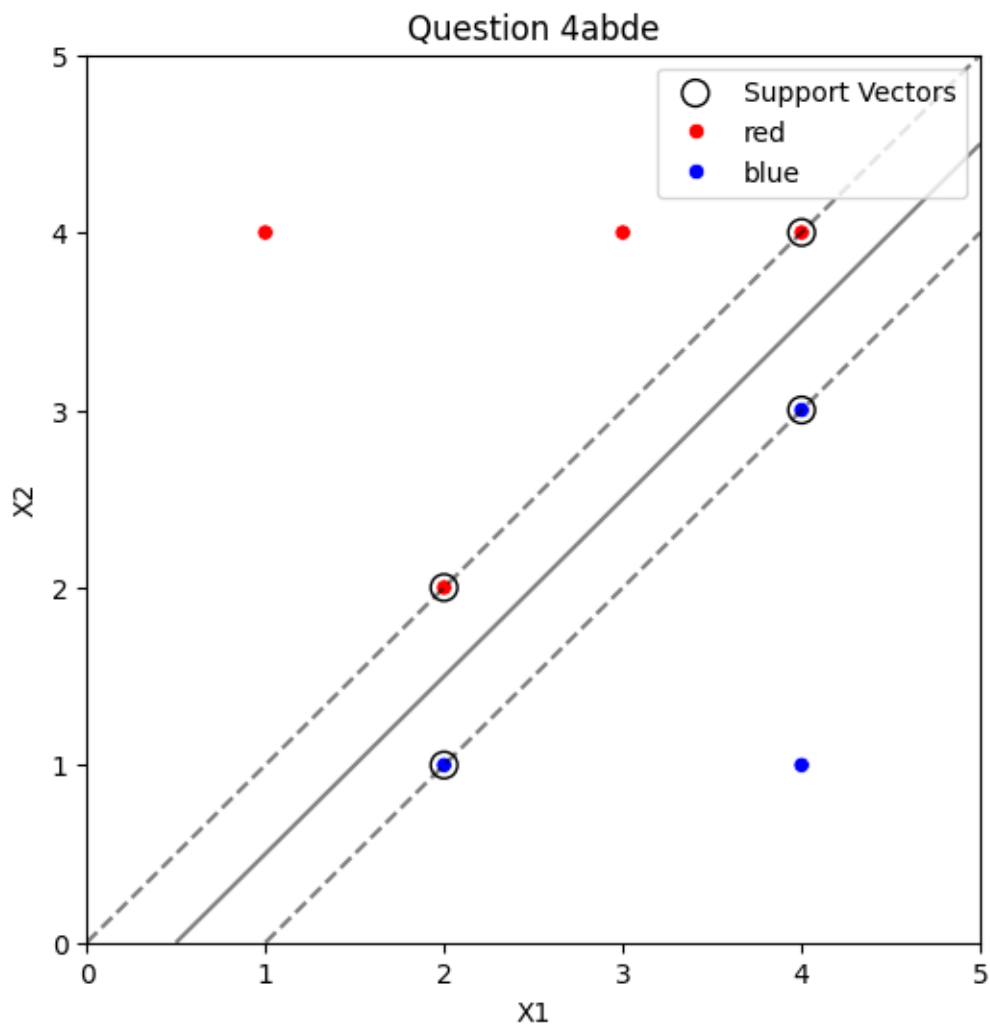


- b. The set of points for which $(1 + X_1)^2 + (2 - X_2)^2 > 4$ are the points outside the circle, whereas the set of points for which $(1 + X_1)^2 + (2 - X_2)^2 \leq 4$ are the points on and inside the circle.

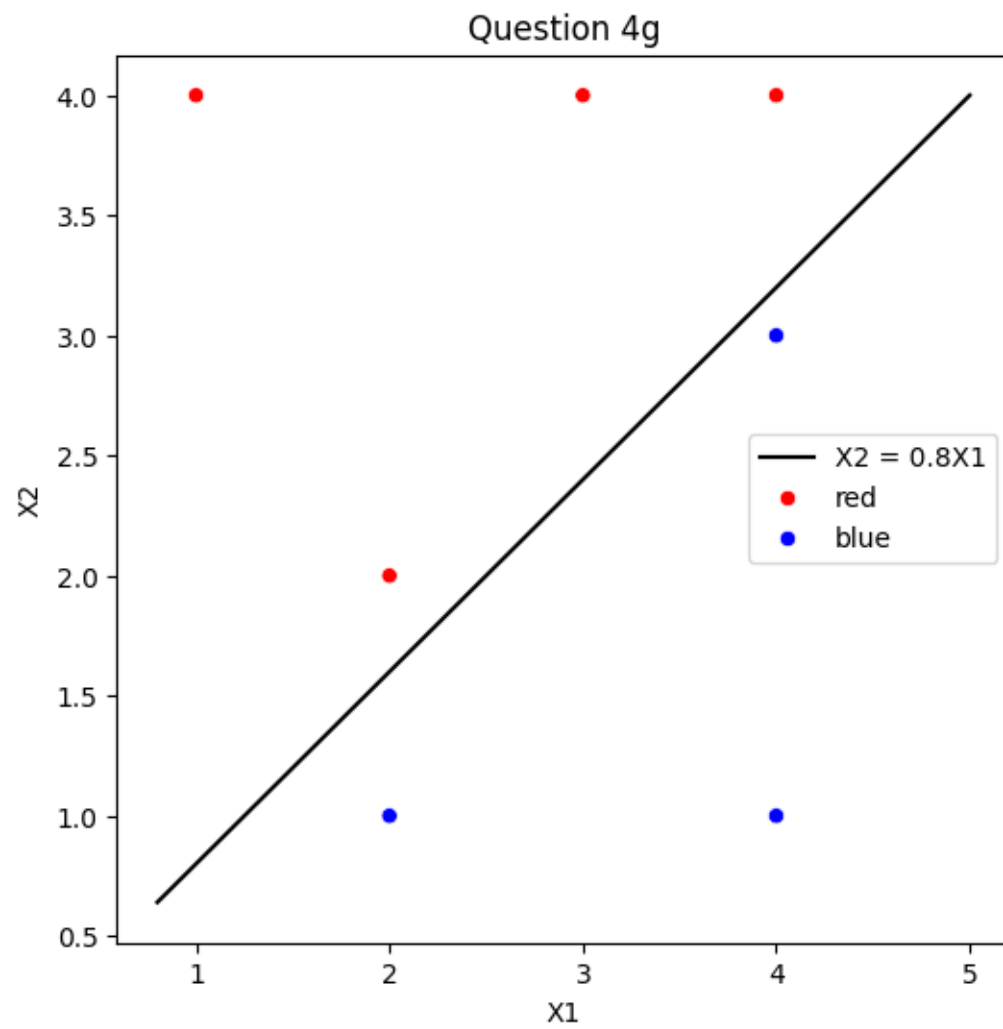
Points	$(1 + X_1)^2 + (2 - X_2)^2$	Class
(0,0)	5	Blue
c. (-1,1)	1	Red
(2,2)	9	Blue
(3,8)	52	Blue

- d. If we expand the equation for the decision boundary to $5 + 2X_1 - 4X_2 + X_1^2 + X_2^2 = 4$, we get an equation that is linear in terms of X_1, X_2, X_1^2, X_2^2

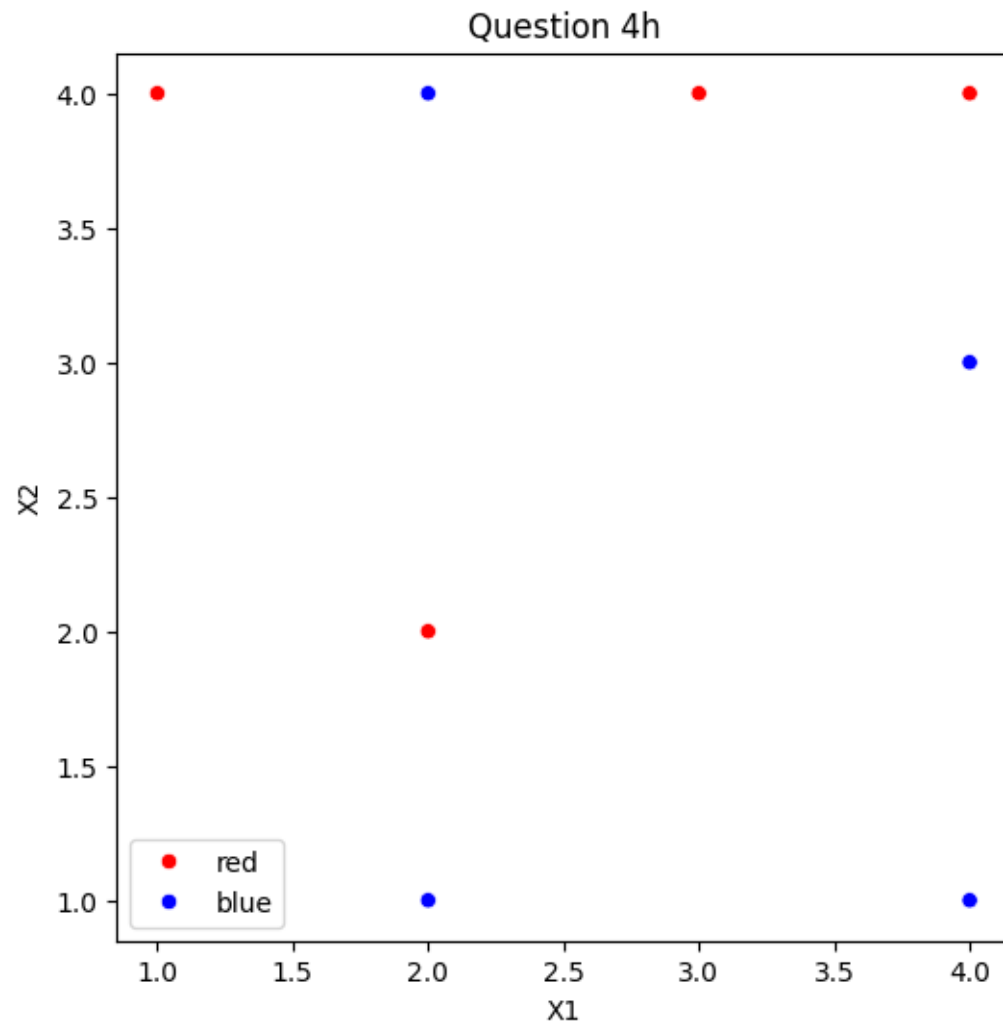
Question 4



- c. Classify to Red if $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$ and Blue otherwise.
In this case, $\beta_0 = -0.5$, $\beta_1 = 1$, $\beta_2 = -1$
- f. The 7th observation $(4, 1)$ if moved slightly will not affect the hyperplane because it is not a support vector, the equation of the hyperplane does not depend on it.
- g. The equation for this hyperplane is $0 = 0.8X_1 - X_2$

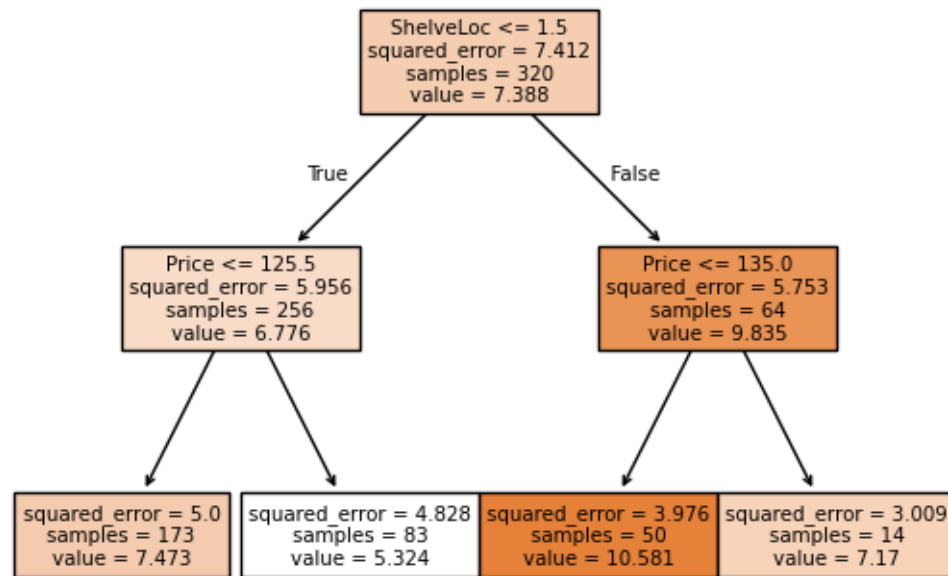


- h. A point that would make the two classes unseparable by a hyperplane would be (2,4) with the category Blue.



Question 5

b. The mean squared error is 6.243



c. Skipped

d. Skipped

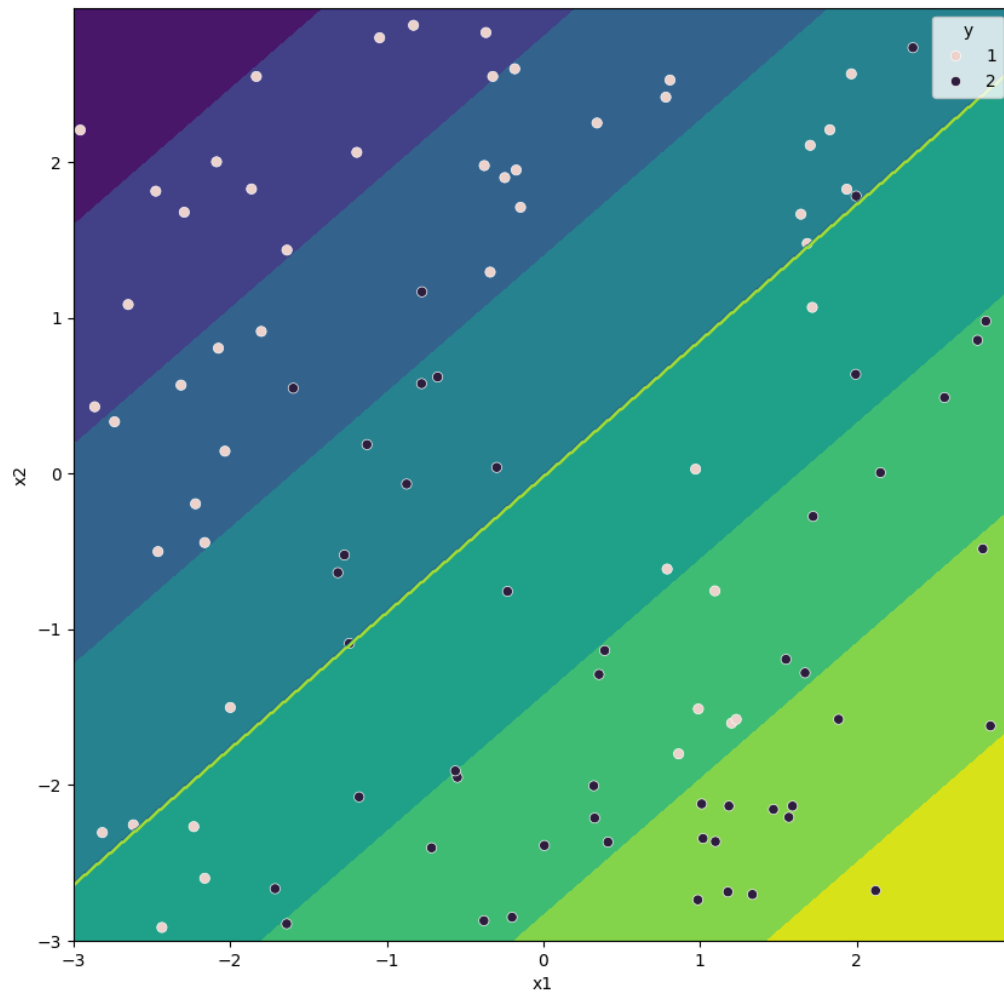
e. Skipped

Question 6

Linear kernel support vector machine

Training score: 0.77

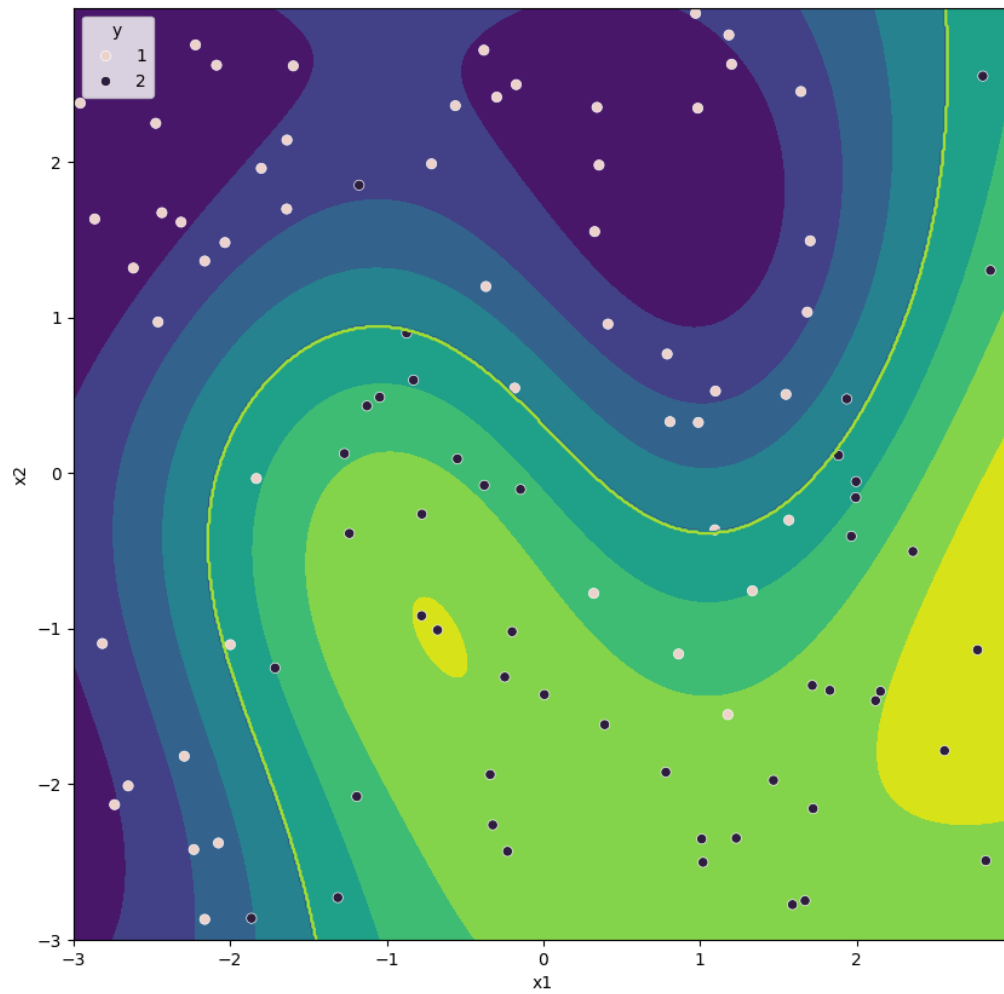
Test score: 0.72



Radial kernel support vector machine

Training score: 0.89

Test score: 0.79



Therefore, radial kernel works better in this case of non-linear separation between the two classes.